Comparative Genomics:
Cis Regulatory Evolution

Assignment 13

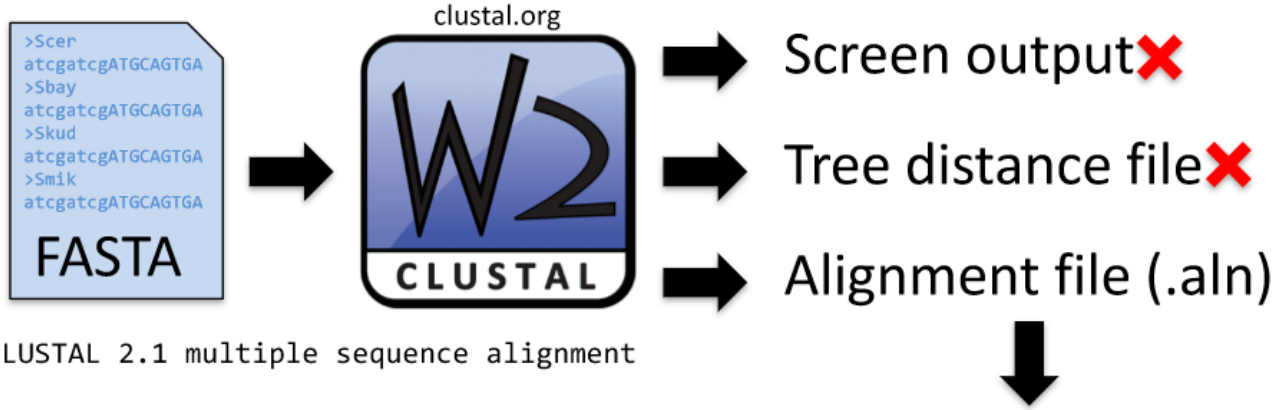
Bio5488 Genomics

04/17/2026

Comparative genomics

- Find conserved promoter sequences in four yeast species and look for transcription factor binding sites
- Outline:
  - Using clustalw to align sequences
  - How much conservation is expected (null model)?
  - Third base wobble positions
  - JASPAR database

Clustalw: multiple sequence alignment

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CLUSTAL 2.1 multiple sequence alignment


Skud            --TTATTTTTTTGCAGCCAACTTCACTACAGTTTAACCTACCTTATAGAAAGAGGTGAAA
Smik            ------TTTTTAACATCTATCCTTGATGA----TAGCATGTTAT-TACGAAATAATACAA
Scer            --TACTTTTTTGATGGCTACAATCATAAT---TTGACATACCTCATATGAA--TATGATG
Sbay            TTTTCTTTTTTAAGCGCCGGTCTAATTATCGGTTATCCTGCTACGCCAAAAGTGATATTA
                      *****     *     *          *  * *          **    *

Skud            TAAAGATGGCGTAAGACAAGTGGATGATTTCCAGACGCAGTGA
Smik            CAAAGATGGTATAAAACAACTTGATGATTTCCAGACACAGTAA
Scer            TAAAGATGGCATAAGACAAGTAGATGACTTCCAGGCACAGTGA
Sbay            TAAAGACGGTATAAGACAAATAGATGACTTCCAGACAAAATGA
                 ***** **  *** **** * ***** ****** *  * * *
```

Identify promoter regions that are more conserved than expected

1) What is the expected rate of conservation at a neutrally evolving site in the genome?

- Third base wobble position

Second letter

First letter	U	C	A	G	Third letter
U	UUU, UUC Phe UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr UAA Stop UAG Stop	UGU, UGC Cys UGA Stop UGG Trp	U C A G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U C A G
A	AUU, AUC, AUA Ile AUG Met	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn AAA, AAG Lys	AGU, AGC Ser AGA, AGG Arg	U C A G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U C A G

Species	Codon	Amino acid
Skud	TGG	Trp
Smik	TGG	Trp
Scer	TGG	Trp
Sbay	TGG	Trp

Only one codon for Trp

Species	Codon	Amino acid
Skud	CAT	His
Smik	CAT	His
Scer	CAA	Gln
Sbay	CAA	Gln

Not same amino acid

Possible 3rd base wobble?

Species	Codon	Amino acid
Skud	GCC	Ala
Smik	GCC	Ala
Scer	GCC	Ala
Sbay	GCC	Ala

Species	Codon	Amino acid
Skud	TTG	Leu
Smik	TTG	Leu
Scer	CTG	Leu
Sbay	CTG	Leu

First two codons differ

Identify promoter regions that are more conserved than expected

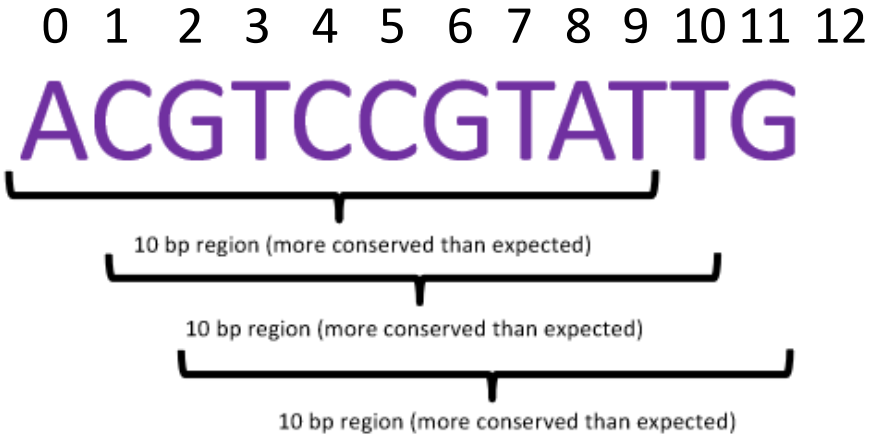
1) What is the expected rate of conservation at a neutrally evolving site in the genome?

- Third base wobble position

2) How many conserved bases out of ten would be necessary to observe significantly more conserved bases than expected?

- Binomial distribution, scipy.stats module (python script not necessary)

Overlapping windows – Report entire sequence



Column	Description	Example
0	Start Position (0-based)	0
1	Stop Position (0-based)	11
2	Scer Sequence	ACGTCCGTATTG

WATCH OUT!

- python indexing = string[0:12]
- stop position = **11**

https://jaspar.elixir.no/collection/core/

database

JASPAR2020

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Search profile(s)

Home > search

Search JASPAR database... Search

Examples: SPI1, P17676, ChIP-seq, Homo sapiens

Advanced Options

184 profile(s) found

Display 10 profiles

Filter:

ID	Name	Species	Class	Family	Logo
MA0265.1	ABF1	Saccharomyces cerevisiae			
MA0266.1	ABF2	Saccharomyces cerevisiae	High-mobility group (HMG) domain factors		
MA0267.1	ACE2	Saccharomyces cerevisiae	C2H2 zinc finger factors	Other factors with up to three adjacent zinc fingers	
MA0268.1	ADR1	Saccharomyces cerevisiae	C2H2 zinc finger factors	Other factors with up to three adjacent zinc fingers	
MA0269.1	AFT1	Saccharomyces cerevisiae			
MA0270.1	AFT2	Saccharomyces cerevisiae			
MA0271.1	ARG80	Saccharomyces cerevisiae	MADS box factors		
MA0272.1	ARG81	Saccharomyces cerevisiae	C6 zinc cluster factors		

Analyze selected profiles

Please select matrix profiles on the left side to add to your cart or perform the following analysis.

- Add to cart
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- Add to cart | View cart
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- Randomize
- Permute
- Download

High quality transcription factor binding profile database

What to turn in:

- A commented 'neutral_rate.py`
- Your clustalw output file PRE1.aln
- Your neutral_rate.py output file S_cer_conserved.txt
- Your README.txt with the answers to all questions and the commands you used to answer the questions